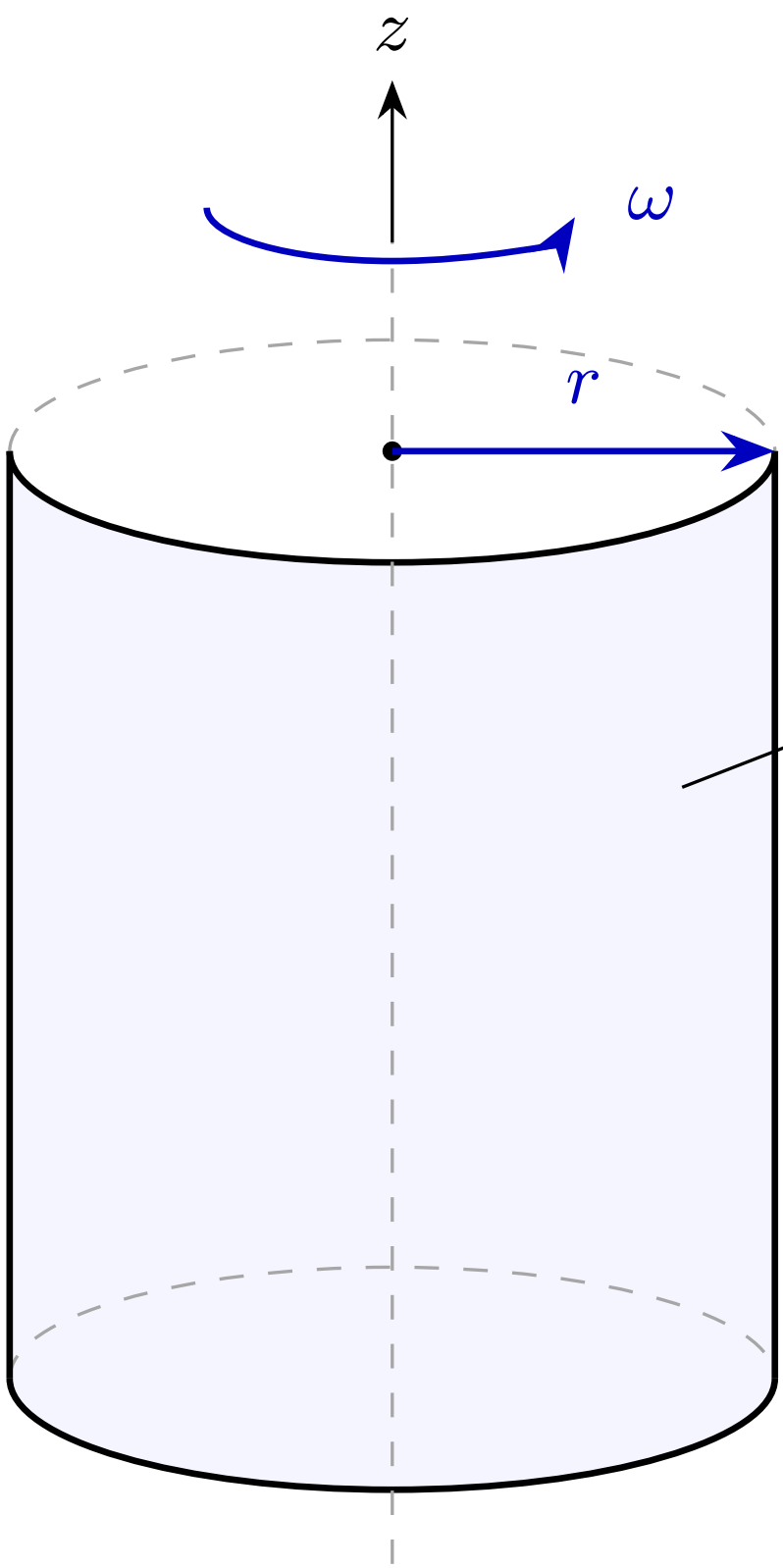
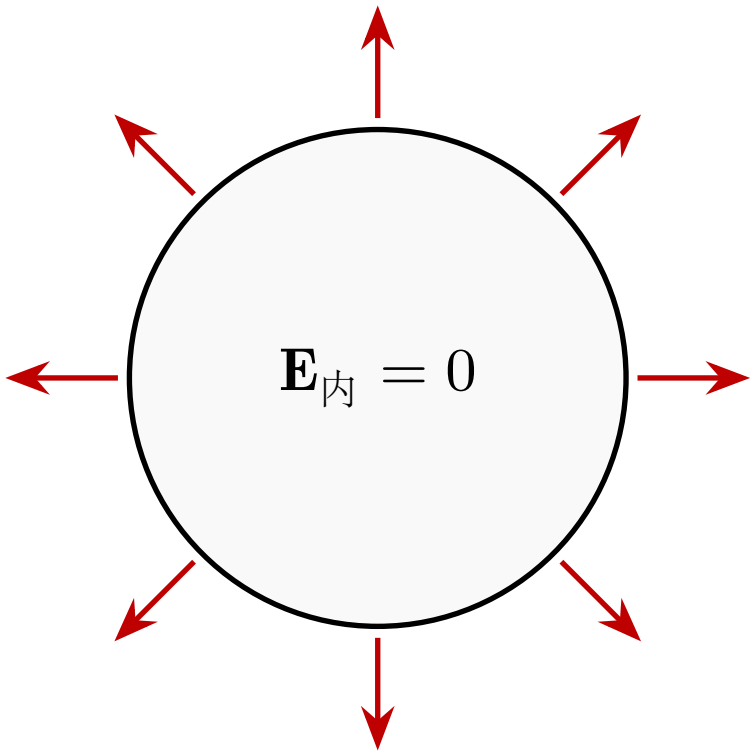


σ：面电荷密度

μ：总面质量密度

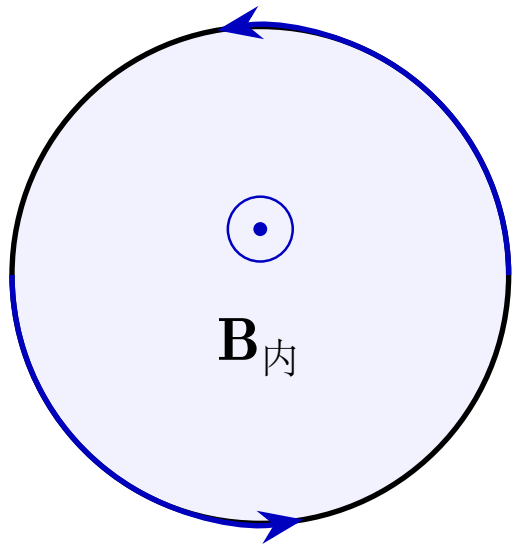
从 $+z$ 看，图示取 $\sigma > 0,\ \omega > 0$

电场



$$\mathbf{E}_{外} = \frac{\sigma}{\varepsilon_0}\mathbf{e}_r$$

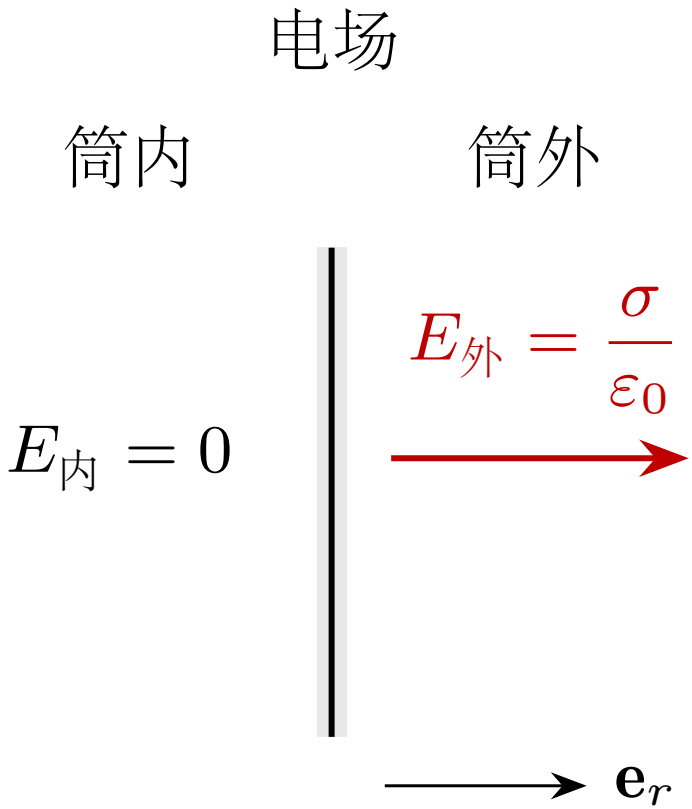
面电流与磁场

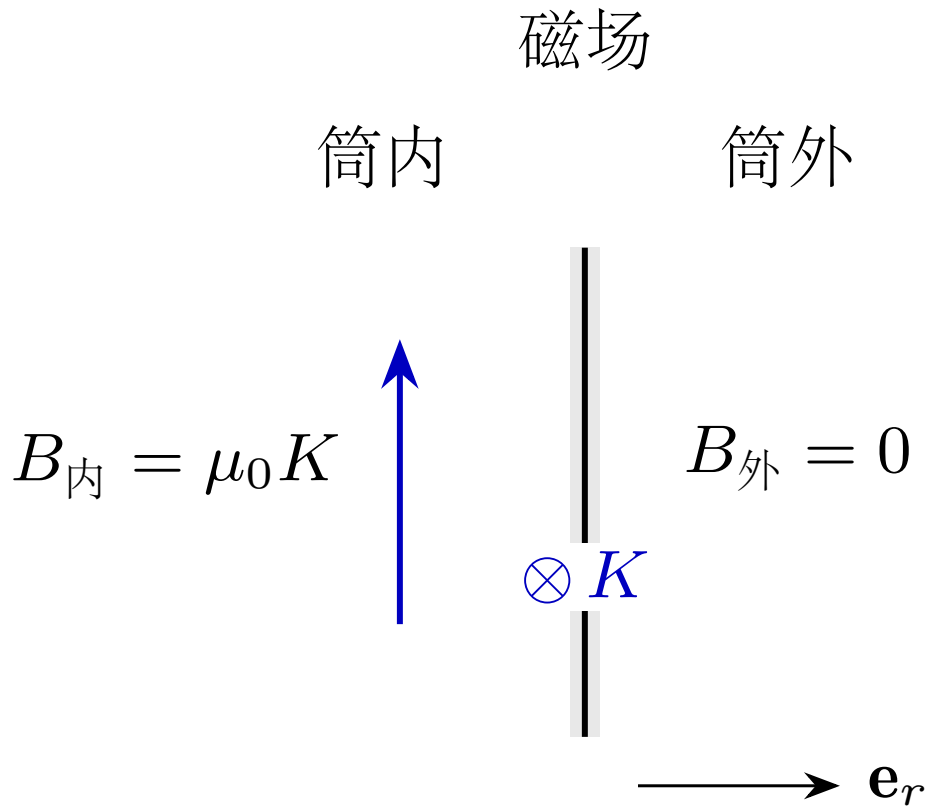


$\mathbf{B}_{外} = 0$

$$\mathbf{B}_{内} = \mu_0 \sigma \omega r\, \mathbf{e}_z$$

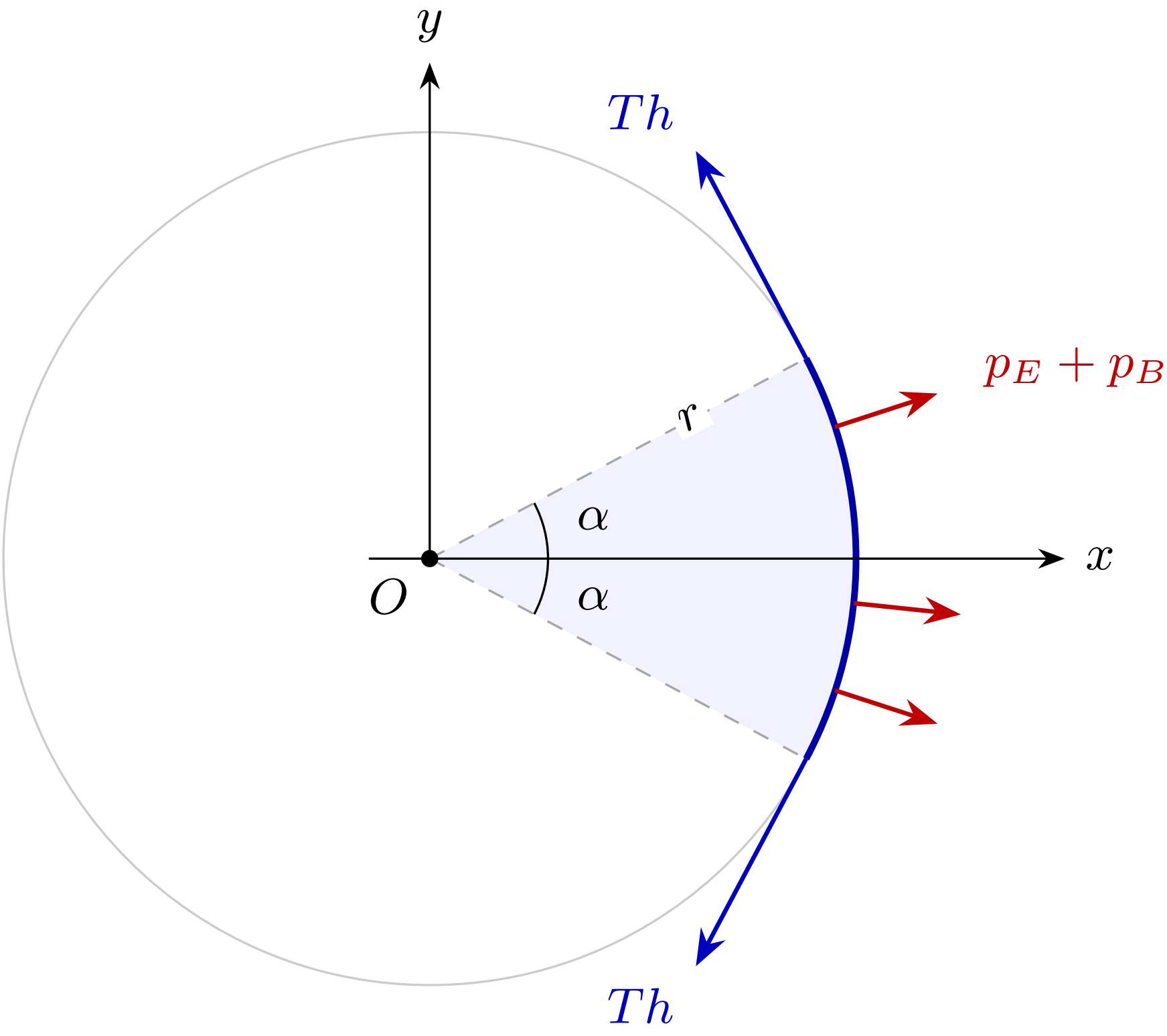
局部截面；$\mathbf{e}_z$ 向上，$\mathbf{e}_\varphi$ 朝纸内



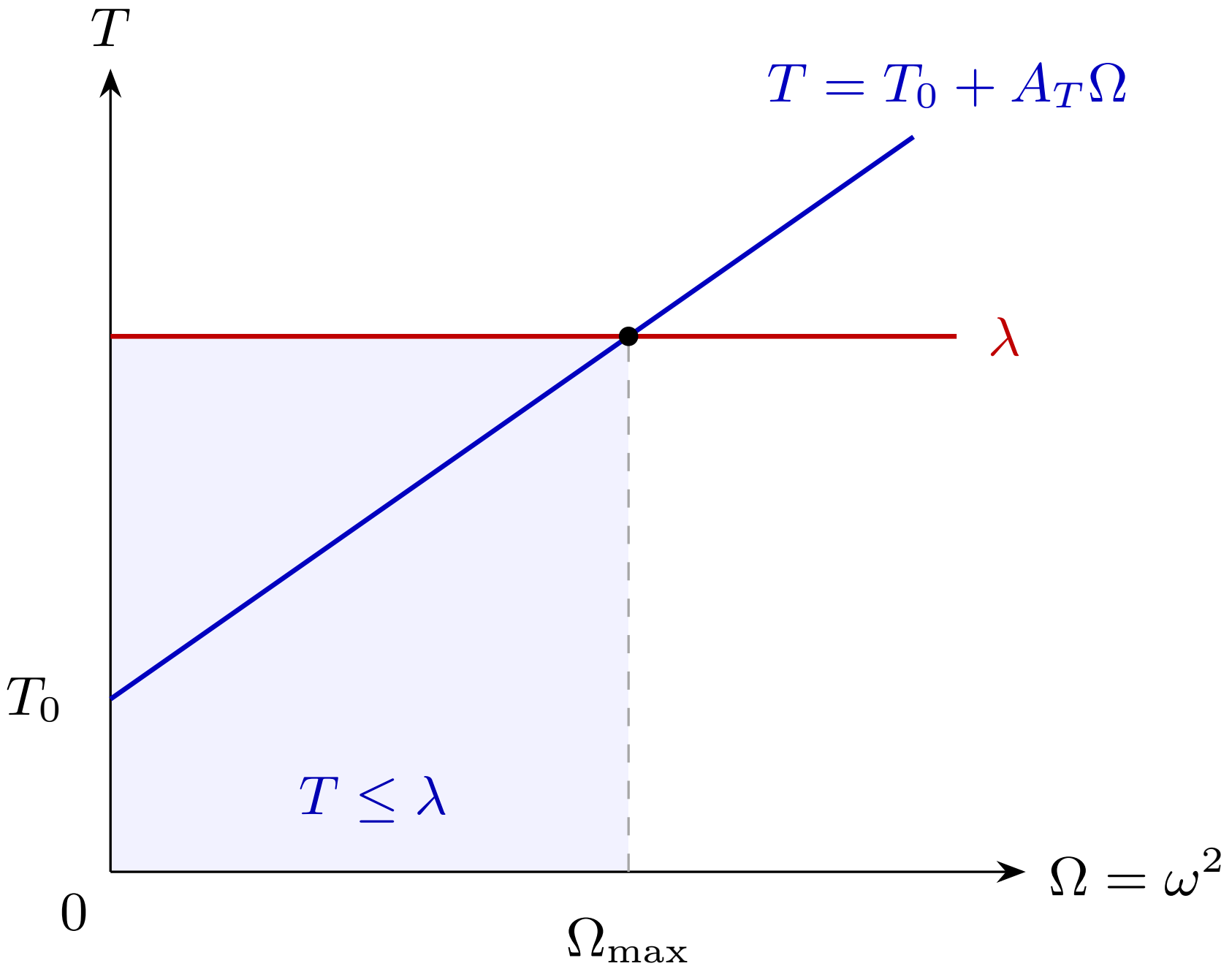


$$\mathbf{E}_{有效}=\frac{\sigma}{2\varepsilon_0}\mathbf{e}_r$$

$$\mathbf{B}_{有效}=\frac{\mu_0 K}{2}\mathbf{e}_z$$



壁带轴向长 h；张力沿两端切线


T
$T = T_0 + A_T\Omega$
λ
T_0
$T \leq \lambda$
$\Omega = \omega^2$
0
$\Omega_{\max}$